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\documentclass{article}
\usepackage{amsmath}
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\begin{document}
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```
\section*{Cramér's V}
```

```
\[
V = \sqrt{\frac{\chi^2}{N \cdot (k - 1)}}
\]
```

```
\noindent
```

```
\textbf{Where:}
```

```
\begin{itemize}
```

```
  \item  $\chi^2$ : Chi-square statistic from a contingency table
```

```
  \item  $N$ : Total number of observations
```

```
  \item  $k$ : The smaller of the number of rows or columns in the contingency table
```

```
  \item  $k = \min(\text{number of rows}, \text{number of columns})$ 
```

```
\end{itemize}
```

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\end{document}
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Putting it all together, the relative change in the soft assignment probability is approximated as:

```
\[
\frac{q_{ij}^{\text{new}} - q_{ij}}{q_{ij}} \approx \frac{\alpha + 1}{2\alpha} \left( 1 + \frac{z_i - \mu_j}{\sqrt{2\alpha}} \right) \left[ 2(z_i - \mu_j)^T \Delta \mu_j - \|\Delta \mu_j\|^2 \right]
\]
```

where:

```
\[
\Delta \mu_j = \frac{m}{N - m} (\mu_j - \bar{z}_S)
\]
```

Here:

```
\begin{itemize}
```

```
  \item  $z_i$ : embedding of sample  $i$ 
```

```
  \item  $\mu_j$ : original centroid of cluster  $j$ 
```

- \item $\bar{z}_S$: mean of the removed points from cluster j
- \item m : number of removed points from cluster j
- \item N : original number of points in cluster j
- \item α : degrees of freedom for the Student's t-distribution

\end{itemize}

\end{document}

\documentclass{article}

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\begin{document}

\section*{Score Formulas}

\subsection*{Score 1}

[

\text{Score of } A = \beta_{A\hat{Z}} = V(\chi^2_{A\hat{Z}})

]

\subsection*{Score 2}

Let cluster be $\{A, B, C\}$

[

\text{Association within the cluster of } A = \Pr(A \in \{A, B, C\}) \times (\frac{\beta_{AA} + \beta_{AB} + \beta_{AC}}{3}) \times \beta_{A\hat{Z}}

]

[

\text{Score of } A = \text{Association within Cluster}

]

\subsection*{Score 3}

Let cluster be $\{A, B, C\}$

[

\text{Association outside the cluster of } A = \left[1 - \Pr(A \in \{A, B, C\})\right] \times \left[\frac{\text{Avg}(\beta_{AA}, \beta_{AB}, \beta_{AC})}{\hat{\chi}^2(\{A, B, C\})}\right]

]

[

\text{Score of } A = (\frac{\text{Association within the cluster of } A}{\text{Association outside the cluster of } A})

]

\end{document}